Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Jonathan Rogers	**Email**	Jonathan.rogers@ae.gatech.edu
Institute Name	Georgia Institute of Technology			
Address	756 W Peachtree St NW, Atlanta, GA 30308			

2. Technology Overview

Title of Proposed Technology	*Design, Prototyping, and Flight Test of a Drone Interceptor for Counter-UAS Operations*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity (√), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and materials		*Mechanical and materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Aerospace Systems	*Fixed-wing/rotary-wing Aircraft mechanical systems*	*Robotics / Automation machinery*	*Robot Control and Intelligence Technologies*
Concept of Proposed Technology	The proposed technology is a drone "interceptor" designed to capture another drone, for the purposes of counter-UAS operations. Counter-UAS operations seek to identify and, oftentimes, destroy drones that are operating in unauthorized areas such as near airports, stadiums, etc. These types of unauthorized drone incursions are becoming more commonplace throughout the world. Many counter-UAS systems (such as "soft-kill" devices) cause the drone to crash. This can be undesirable if, for instance, it is flying over a runway or a crowd of people. We have developed a			

(max 10 pages)

	"drone interceptor" designed to autonomously fly to the hostile drone and capture it in a basket. The purpose of this system is to capture the drone while in flight, and then bring it to a location where it can be inspected and investigated further. The main components and functions of the system are: • A high-performance drone equipped with a capture basket, capable of autonomously flying to and capturing the target drone. • The interceptor will have specialized guidance and control algorithms as well as onboard sensing for localizing and intercepting the drone with low relative velocity. • The interceptor would be designed to have higher acceleration performance than most commercial drones, meaning it would be capable of intercepting targets which are actively maneuvering. The system would be designed for deployment near high-value infrastructure such as airports, stadiums, power plants, etc.
Rationale for Joint R&D Project	The proposed project requires close collaboration between a research institute (such as Georgia Tech) and a commercial partner. The research lab will focus on developing several challenging underlying components including vehicle guidance algorithms and sensors / algorithms for target vehicle localization. The commercialization partner will develop the vehicle design, integrated system, and user interface which will be essential for real-world deployment. Rationale for collaboration: • The research institution will provide expertise in air vehicle guidance algorithms, sensing (such as radar, visual/IR cameras, and/or acoustics) for identifying and localizing the target drone, and preliminary vehicle sizing to ensure maneuver overmatch with respect to the target. • The Korean SME will provide expertise in system design and integration, user interface design, and flight experimentation / testing. • Close collaboration between the researchers at Georgia Tech and product development team at the Korean SME will ensure that the vehicle is capable of reliable operation and meets performance needs for counter-UAS missions.
Key Words	1. Counter-UAS / Counter-drone 2. UAS guidance and control 3. Aerial manipulation

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]		**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives		The overall R&D objectives of the project are to 1) develop the component technologies needed to build a drone interceptor for small UAS in-flight capture, and 2) build and test a prototype drone interceptor in an outdoor environment. The specific R&D objectives are as follows: 1) **Develop interceptor guidance and control algorithms.** The interceptor will need to intercept the target drone with minimal relative velocity in order to capture it successfully. The team will identify (or modify, as necessary) aerial intercept algorithms from the literature to use in the prototype system. 2) **Develop sensor package and sensor processing algorithms.** The team will identify a suitable sensor package that can detect the drone at ranges out to 200 m using, potentially, a combination of radar, LIDAR, camera, and/or acoustic sensors. The team will develop sensor fusion algorithms to continuously estimate the position of the target with respect to the interceptor during the capture sequence. 3) **Develop vehicle design.** The team will design the vehicle with sufficient thrust and maneuverability to provide a maneuver overmatch capability against typical small UAS. The vehicle will include a capture basket on top of the aircraft (prior lab experiments [1] have shown the benefit of mounting the capture basket above rather than below the vehicle due to downwash effects). 4) **Build an integrated prototype drone interceptor vehicle.** The project team will construct a prototype drone interceptor, including the identified sensor package and the guidance, control, and sensing algorithms developed during the project. A basic user interface will also be developed for high-level command and control. 5) **Experimentation and aerial capture demonstration.** The project team will flight test the aircraft in an outdoor environment. A progression of flight tests will lead to eventual demonstration of autonomous aerial capture of a small, uncooperative UAS in an outdoor environment.			
R&D Target Specifications		Target localization (>90%): The prototype interceptor drone will be able to identify and maintain a continuous track estimate of the target small UAS greater than 90% of the time, as long as the target remains within 200 m of the interceptor. Aerial capture rate – non-evading target (>90%): The prototype interceptor drone will successfully capture the target drone autonomously greater than 85% of the time, when the target is not actively trying to evade the interceptor. Aerial capture rate – evading target (>70%): The prototype interceptor drone will successfully capture the target drone autonomously greater than 65% of the time, when the target is actively trying to evade the interceptor.			

(max 10 pages)

		Target localization accuracy (< 1 m when within 50 m of interceptor, < 1 cm when within 5 m of interceptor): The interceptor will estimate the position of the target drone to better than 1 m accuracy when within 50 m of the interceptor, and better than 1 cm accuracy when within 5 m of the interceptor. Interceptor maneuverability (> twice max maneuver acceleration of typical small UAS): The prototype interceptor drone will have over twice the maximum acceleration capability of typical small, commercially-available UAS.
R&D Plan		The proposed project will be completed over three years. Year 1 is focused on developing and validating the component technologies in simulation and laboratory experiments. Year 2 is focused on full prototype construction and flight testing in indoor environments, to include simulated and actual drone captures. Year 3 will focus on outdoor flight testing, characterization of the localization algorithms, and demonstration of outdoor, autonomous aerial captures. **Year 1: Development and Validation of Component-Level Technologies** • **Task 1.1: Identification of suitable aerial intercept algorithm from literature.** Numerous guidance and path planning algorithms exist for guiding an autonomous aircraft to an object with zero relative terminal velocity – such algorithms were originally developed for aerial refueling or aerial rendezvous. In this task, the team will identify suitable algorithms from the literature, modify them if necessary, and simulate performance to ensure robustness across a wide range of geometries and scenarios. • **Task 1.2. Creation of sensor package for target localization.** The research team will create a sensor package capable of localizing the target drone to a distance of 200m from the interceptor. The team will consider the use of onboard radar, LIDAR, camera(s), acoustic sensors, and potentially other sensors to detect the target. The sensing package will be determined based on the SWAP and cost of sensors that are commercially available. • **Task 1.3: Develop sensor fusion algorithm for target localization.** The research team will use an Extended Kalman filter (or other appropriate estimator) to fuse data from the selected sensors and produce an estimate of the target drone position and velocity with respect to the interceptor. The algorithm will be prototyped and tested in simulation. • **Task 1.4: Design interceptor drone structure and capture mechanism.** The research team will design the drone interceptor vehicle to include the structure, selection of motors/ESCs, and design of the capture basket mechanism and actuator. As a starting point, the team may use design concepts employed on Georgia Tech's proof-of-concept interceptor vehicle described in Reference [1]. • **Task 1.5: Development of basic user interface software.** The research team will develop a basic user interface that allows the operator to maintain situational awareness of the drone as it performs a capture operation. This interface will show the positions of both vehicles and the anticipated trajectories, as well as buttons for basic commands. **Year 2: System Integration and Indoor Flight Testing** • **Task 2.1: Fabrication of interceptor prototype.** The research team will construct a prototype of the interceptor drone using relevant fabrication

		methods. This will include selection and installation of an autopilot (e.g., CubePilot or similar), sensors (selected in Task 1.2), and algorithm implementation on an embedded computer. • **Task 2.2: Indoor flight experiments.** The research team will flight test the interceptor in an indoor environment – for instance, the Georgia Tech Indoor Flight Lab or another space available to the Korean SME. Once basic vehicle performance is established, aerial captures will be attempted indoors. Target localization may be performed either using the onboard sensors or offboard sensing (such as motion capture) in these preliminary tests. • **Task 2.3: Design refinements based on flight testing.** The research team will refine the mechanical, sensor, and/or algorithm designs based on the outcome of these tests. • **Task 2.4: Test localization sensors and algorithms in outdoor environment.** The research team will perform preliminary tests characterizing the performance of the interceptor-based sensor suite for target localization in an outdoor environment. Performance will be compared with the desired accuracy levels and changes to the sensors and/or localization algorithm will be made as necessary. **Year 3: Outdoor localization and capture testing, preparations for technology commercialization.** • **Task 3.1: Outdoor capture testing – non-maneuvering target.** The research team will perform a series of tests in which the interceptor drone finds (localizes), travels to, and captures a hovering drone. These tests will be performed repeatedly from different initial conditions to assess the capture accuracy. Adjustments to the design will be made based on these tests to improve performance as necessary. • **Task 3.2: Outdoor capture testing – maneuvering target.** The team will perform additional capture tests in which the target is flying in a straight line. Tests will be performed at different flight speeds, characterizing the capture rate in different conditions. The tests will conclude with attempts to capture a target that is performing mild evasive maneuvers (e.g., performing turns or climbs as the interceptor approaches). Again, the mechanical, electrical, and software elements will be modified if needed to improve autonomous capture performance based on the observed results. • **Task 3.3: Preparation for commercialization.** The team will assess the commercialization potential of the system based on observed performance, anticipated cost, and potential benefit to customers. The team will schedule calls or briefings with potential customers in the energy and defense sectors to gauge interest. The team may also develop an IP strategy if the commercialization potential is assessed to be promising.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for some of the more fundamental aspects of the work including guidance and localization algorithm development, modeling and simulation of performance, and basic vehicle sizing calculations. The specific roles of the research institute include: • Selection of the guidance algorithm from the literature and tailoring of the algorithm as needed. • Selection of sensors and design of the target drone localization algorithm. • Creating a modeling environment that includes both the interceptor and the target drone, and simulating performance of the guidance and localization algorithms. • Creating a vehicle sizing tool that predicts acceleration capabilities based on rotor size, motor characteristics, and vehicle mass and inertia. This will be provided to Korean SME for use in vehicle design. Resources to be provided: • Computing resources for modeling and simulation tasks • Fabrication labs for component prototyping, as needed • Indoor Flight Lab for indoor experimental testing Research tasks: • The research institute will lead Tasks 1.1, 1.2, 1.3, 2.2, and 2.3. • The research institute will collaborate either equally or in a supporting role with the Korean SME on the remaining project tasks.
Expected R&R of Korean SME Partner	The Korean SME will be responsible for aspects of the work related to system integration, large-scale testing, performance characterization, and preparation for commercialization. The specific roles of the Korean SME include: • Interceptor mechanical design and prototype construction • User interface design and UX/UI software development • Integration of the prototype to include mechanical, electronic, and software components • Flight testing of the prototype to characterize autonomous flight performance and also assess maneuverability with respect to performance goals • Testing in outdoor (and potentially indoor) environments to characterize capture performance and assess against performance goals • Development of IP plan and outreach to potential customers (customer outreach will also be facilitated by research institution) Resources to be provided: • Fabrication facilities (3D printing, electronics integration, etc.) • Small-scale outdoor flight test area Research tasks: • The Korean SME will lead Tasks 1.4,1.5, 2.1, 2.4, 3.1, 3.2, and 3.3. • The Korean SME will collaborate either equally or in a supporting role with the research institution on the remaining project tasks.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed project is expected to lead to a first-of-its-kind drone interceptor designed for counter-UAS operations. The project will develop the needed sensing systems and control algorithms to ensure that small UAS can be captured reliably and autonomously by the interceptor. The R&D tasks will result in experimentally-proven algorithm(s), sensor suite designs, and mechanical designs for the interceptor aircraft. An equally important outcome is the production of flight test results, which will illustrate performance against example small UAS targets and provide localization accuracy and capture rates. These metrics will be important as the technology matures toward commercialization. The project is addressing an important need in the defense and energy sectors for systems that can eliminate threats from unauthorized small UAS operating near sensitive areas. While some counter-UAS systems exist, most of them perform “soft kills” wherein the target drone loses its connection with its controller or suffers from sensor or navigation failures and crashes. These types of uncontrolled crashes can be undesirable in sensitive areas or near crowds of people. The proposed interceptor will offer a unique solution that captures the target UAS and relocates it to a safe area for containment. The commercialization potential for such a system could be quite significant. In addition, the successful fielding of such a system at scale could substantially improve security against small UAS threats and protect critical national infrastructure. The Korean SME will benefit from the proposed collaboration in that the research institution will provide important expertise in the areas of guidance algorithms, sensor fusion and estimation, and drone simulation and modeling. The research institution has already designed a basic prototype that can perform successful indoor captures of a small target drone, and the lessons learned from this prototype development (documented in [1]) will be brought to bear on the proposed project to increase the chances of success. The overall goal of the project is to support the Korean SME to ensure that the project sets the stage for a **viable commercial product.** As such, a critical outcome of the proposed work is a substantial set of test results demonstrating successful performance in real-world settings.

6. References

(1) T. Bumbary, “Development of a UAV-Mounted System for Physical Interception of Aerial Vehicles,” Master’s Thesis, Georgia Institute of Technology, Atlanta, GA, 2025.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization